

# Which Issues Divide?

Interpretable Ideal-Point Estimation from Political Text

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Jefferson Leal

September 2, 2026

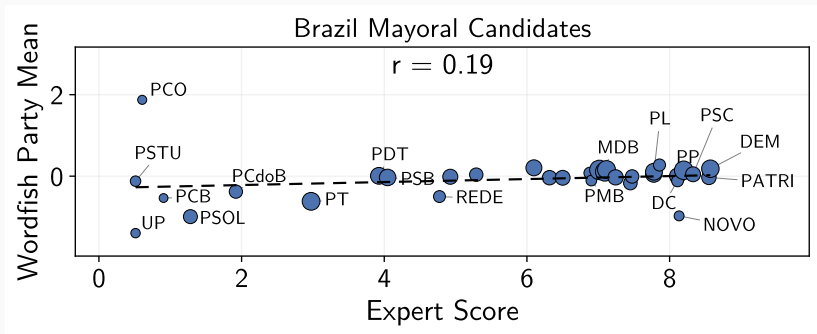
University of Rochester

# When Scaling Ideology from Text Goes Wrong

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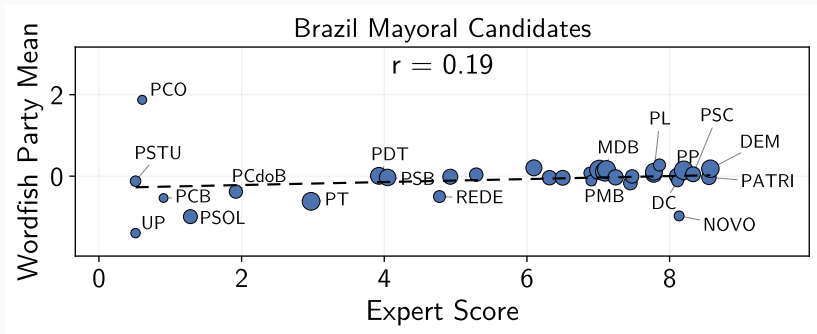
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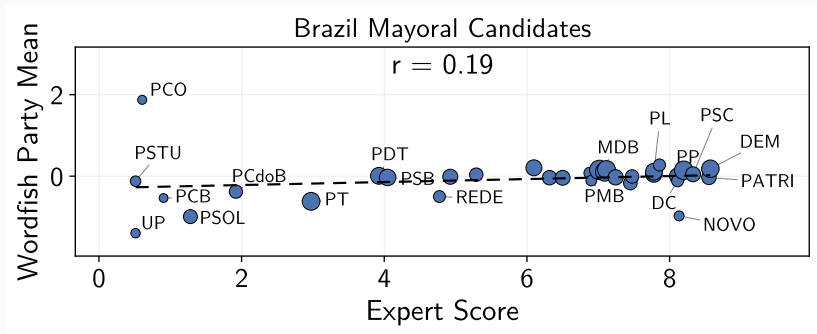
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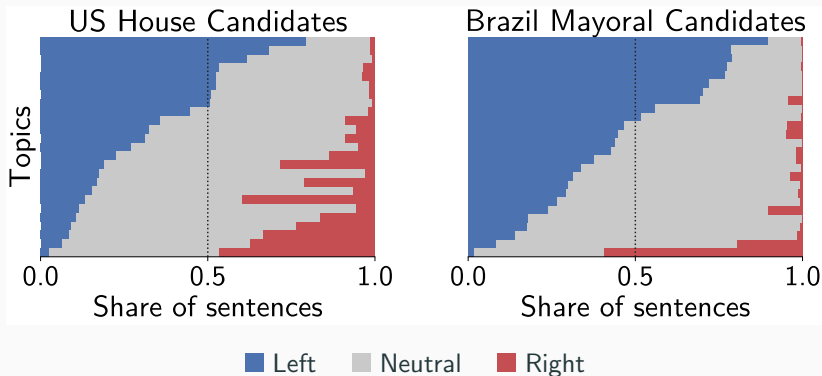
- Supervised learning? divides must be known in advance.
- Deep learning or LLMs? Black boxes.

# Problem: Non-Polarized Language

- **Data:** 4,507 US House platforms, 2018–2022 primaries ([Porter et al., 2025](#)); 16,830 Brazilian mayoral platforms, 2020 ([Tribunal Superior Eleitoral, 2020](#)).

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# Problem: Non-Polarized Language

- Global South: valence issues, i.e., public services & corruption control ([Kitschelt, 2007](#); [Bleck and van de Walle, 2012](#)).
- Local politics in the US is similar ([Bucchianeri, 2020](#)).

- Candidates reveal ideology through what issues they emphasize (**salience**) and the stated position per issue (**stance**) ([Laver, 2001](#); [Lauderdale and Herzog, 2016](#)).

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# Fasti Pipeline

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## Two Ideas: Emphasis and Stance

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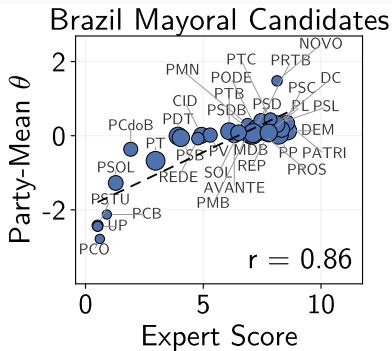
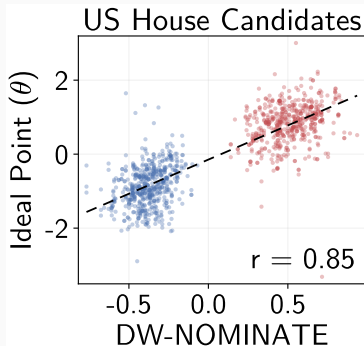
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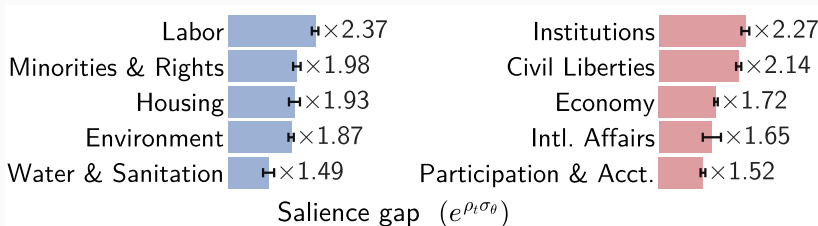
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- One ideal point per candidate generates both.
- **Stance**: a linear latent factor model on ideal point.
- **Salience**: counts are multinomial across topics, each topic's expected share quadratic in the ideal point.

# External Benchmarks

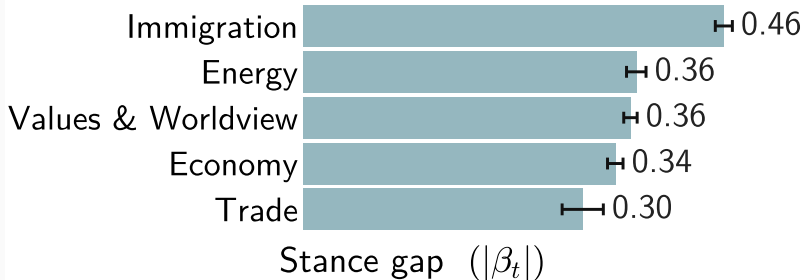


# Salience Divisions: United States

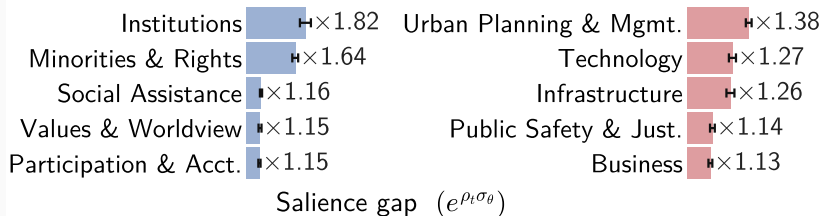


■ Left-emphasized    ■ Right-emphasized

## Stance Divisions: United States

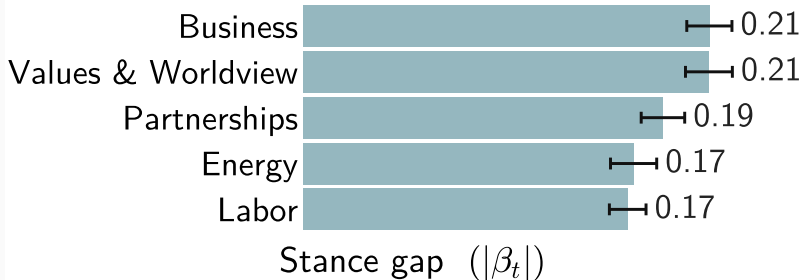


# Salience Divisions: Brazil



■ Left-emphasized    ■ Right-emphasized

## Stance Divisions: Brazil



# Candidate Strategic Positioning

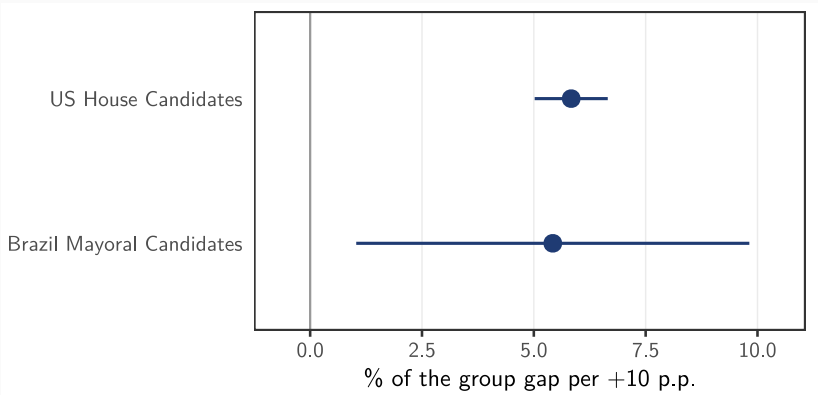
- Candidates strategically reposition to match the leaning of the district on presidential elections ([Meisels, 2026](#)).

# Candidate Strategic Positioning

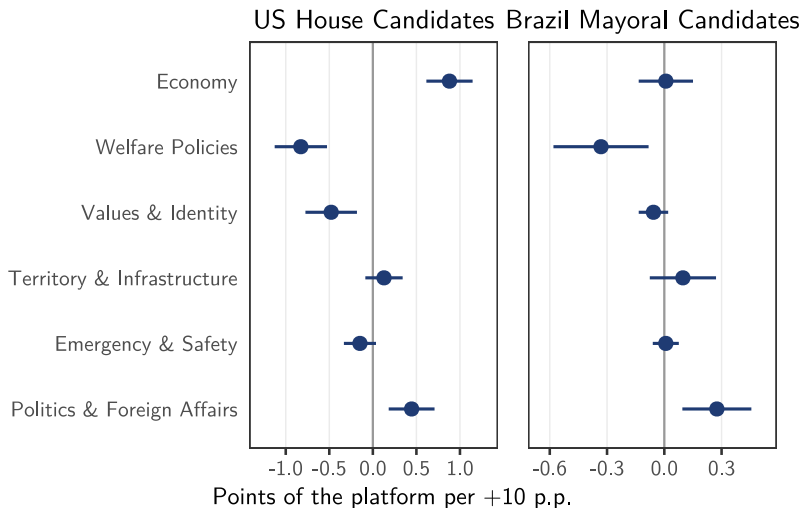
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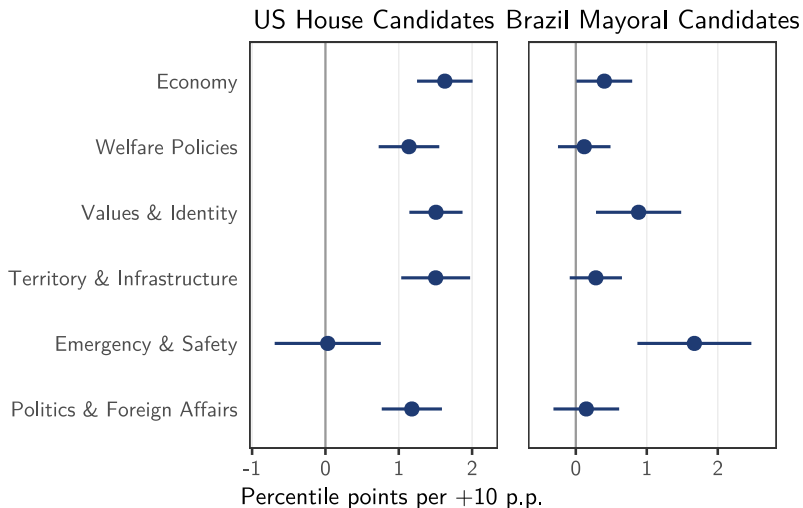
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# Issue Salience Positioning



# Issue Stance Positioning



- **Interpretability gain:** Fasti identifies which issues divide, by salience and by stance, so construct validity is easy to assess.

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- **LLM caveat:** minor prompt variations (party and name cues) move extreme candidates; implausible divides in Brazil, i.e., international affairs and immigration.

# Conclusion

- **Interpretability gain:** Fasti identifies which issues divide, by salience and by stance, so construct validity is easy to assess.
- **LLM caveat:** minor prompt variations (party and name cues) move extreme candidates; implausible divides in Brazil, i.e., international affairs and immigration.
- Even with LLM advances, the IRT model will remain useful: interpretable parameters; quick estimation.

# Thank you!

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## Related Work

- **Unsupervised scaling:** Wordscores, Wordfish ([Laver et al., 2003](#); [Slapin and Proksch, 2008](#)); embedding-based scaling ([Rheault and Cochrane, 2020](#); [Case, 2025](#)).
- **Topic-wise scaling:** Wordshoal, a factor model over topic-specific positions ([Lauderdale and Herzog, 2016](#)).
- **Behavioral and expert benchmarks:** CFscores, DW-NOMINATE, expert placements ([Bonica, 2014](#); [Bolognesi et al., 2023](#)). External, not text-derived.
- **Stance detection with NLI:** entailment classifiers reach strong accuracy with modest training data ([Laurer et al., 2024](#); [Burnham et al., 2026](#)).
- **Platforms vs. roll calls:** what candidates say is related to, not the same as, how they vote ([Meisels, 2026](#)).

# Classical Text Scaling

- Bag-of-words models e.g., Wordfish.
- **Assumption:** word choice reveals issue salience: 'taxes' read as right, 'environment' as left ([Grimmer and Stewart, 2013](#); [Lowe et al., 2011](#)).
- **Problem:** when sides share vocabulary but take opposite stances, the signal is too subtle to recover ([Lauderdale and Herzog, 2016](#); [Rheault and Cochrane, 2020](#)).
- On less-polarized text, many measurement tools are hard to interpret and have validity issues.

# Wordfish Model

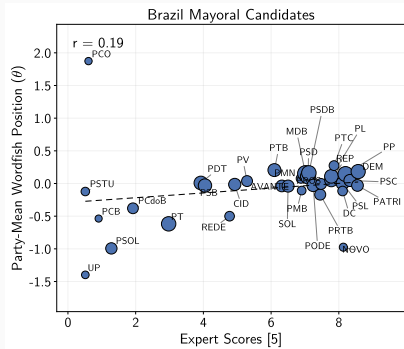
Feature  $w$  in document  $i$ , count  $Y_{iw}$ ; the document's location on the latent scale is  $\theta_i$  (Slapin and Proksch, 2008):

$$Y_{iw} \sim \text{Poisson}(\lambda_{iw}), \quad \log \lambda_{iw} = \alpha_i + \psi_w + \beta_w \theta_i$$

- $\theta_i$ : the recovered scale (identified only up to location and scale).
- $\alpha_i$ : document length;  $\psi_w$ : feature baseline frequency.
- $\beta_w$ : feature weight — pulls a document toward one pole.

Unsupervised:  $\theta_i$  and every  $\beta_w$  are fit from counts alone, with no labels to fix which pole is which. Compare the joint model, where  $\beta_t$  and  $\rho_t$  load on supervised sentence codes.

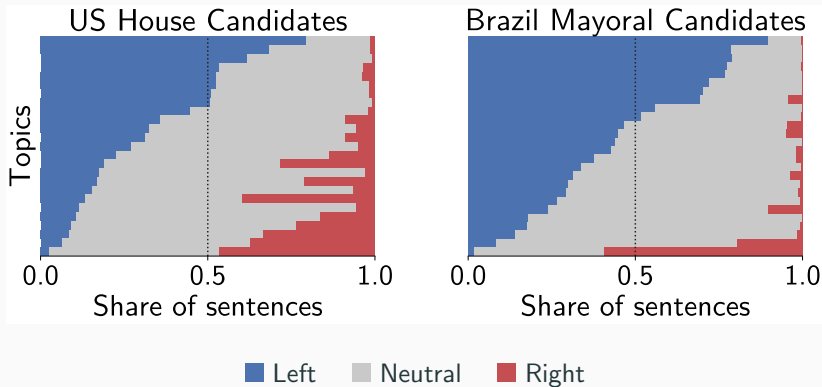
# Wordfish on Brazil: Discretionary Choices



- Party means barely track the experts ( $r = 0.19$ ).
- Discretionary choices have huge influence (Denny and Spirling, 2018): including bi-grams  $r : 0.19 \rightarrow 0.73$ .
- Algorithm only converges if one prunes rare terms (to  $\sim 8,000$ ) and winsorizes counts of frequent terms.

# Corpora

**Corpora:** US Congress campaign platforms, 2018–2022 primaries (CampaignView, [Porter et al., 2025](#)): 4,507 documents. Brazilian mayoral platforms, 2020 elections: 16,830 documents.



# Sentence Segmentation

- One pipeline for both corpora: encoding repair, Unicode normalization, pre-trained spaCy sentence segmenter per language, list splitting, 150-word cap.
- Platform text is list-heavy; off-the-shelf segmentation alone over-splits.
- 439.3K chunks (US) and 3.10M (BR); median 18 and 15 words; median document breaks into 58 and 148 chunks.
- Every chunk is scored and counted. Nothing is discarded at this stage.
- Documents with  $<10$  chunks are dropped from the scale: their mean stance is mechanically  $-1$ ,  $0$ , or  $+1$ . The scale is fit on 4,333 US and 16,674 Brazilian platforms.

# Topic Modeling

- **Pipeline:** sentences  $\rightarrow$  embeddings  $\rightarrow$  topic model  $\rightarrow$  stance  $\rightarrow$  IRT.
- **Sentences:** spaCy + regex rules  $\rightarrow$  chunks  $\leq 150$  words (median 18 and 15 words); 439.3K (US) and 3.10M (BR).
- **Contextual embeddings:** all-MiniLM-L6-v2 (US); paraphrase-multilingual-MiniLM-L12-v2 (BR) ([Wang et al., 2020](#); [Reimers and Gurevych, 2019](#)).
- **Topics:** dimensionality reduction via UMAP  $\rightarrow$  HDBSCAN for clustering ([Grootendorst, 2022](#); [Angelov, 2020](#)).
- **US:** 96 subtopics. **Brazil:** 169 subtopics. Labeled by keywords and examples.
- **Supertopics:** subtopics grouped into 25 (US) and 26 (BR) classes of one codebook; the model estimates over those.

# Topic Model

- **Embeddings:** 384 dimensions. all-MiniLM-L6-v2 (US); paraphrase-multilingual-MiniLM-L12-v2 (BR).
- **Reduction:** UMAP to 10 dimensions ([McInnes et al., 2018](#)).
- **Clustering:** HDBSCAN ([McInnes et al., 2017](#)): density-based, no preset topic count, explicit noise cluster.
- **US:** fit on the full corpus  $\rightarrow$  96 subtopics.
- **Brazil:** fit on 150 platforms per party (4,068 platforms; 770,770 chunks), remaining 2.3M chunks projected through the fitted reduction and cluster membership model  $\rightarrow$  169 subtopics.
- Noise chunks assigned to their highest-probability topic, so every sentence enters  $Y_{it}$ .
- Key terms by normalized class-based TF-IDF.

# Stance Labels

Each sentence takes a stance: **Left**, **Right**, or **Neutral**, defined over two dimensions.

|                | Left                                                                                                | Right                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Economy</b> | State steers the economy: regulation, public services, redistribution, welfare, labour, environment | Market allocates: free enterprise, business incentives, free trade, fiscal orthodoxy, limited welfare |
| <b>Society</b> | Rights, equality, openness: civil rights, democracy & participation, minorities, immigration, peace | Nation, tradition, authority: patriotism, traditional morality, law & order, strong government        |

**Neutral:** no direction. Vague goals, valence and competence claims, anti-corruption, decentralization.

# CMP Crosswalk: Economy

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|                |                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Left</b>    | Market Regulation (403), Economic Planning (404), Corporatism/Mixed Economy (405), Protectionism+ (406), Keynesian Demand Management (409), Controlled Economy (412), Nationalisation (413), Marxist Analysis (415), Anti-Growth/Sustainability+ (416.1–.2), Environmental Protection (501), Welfare State Expansion (504), Education Expansion (506), Labour Groups+ (701) |
| <b>Right</b>   | Free Market Economy (401), Incentives+ (402), Protectionism– (407), Economic Growth+ (410), Economic Orthodoxy (414), Welfare State Limitation (505), Education Limitation (507), Labour Groups– (702)                                                                                                                                                                      |
| <b>Neutral</b> | Economic Goals (408), Technology & Infrastructure+ (411), Culture+ (502), Agriculture & Farmers+ / – (703.1–.2)                                                                                                                                                                                                                                                             |

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+ / – abbreviate Positive/Negative; sub-codes collapse into their parent.

Codes conflating opposite poles (bare 201) are excluded, no guesses.

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# CMP Crosswalk: Society

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**Left**      Anti-Imperialism (103), Military— (105), Peace (106), Internationalism+ (107), Human Rights (201.2), Democracy (202), Pre-Democratic Elites— (305.5–.6), Equality+ (503), National Way of Life— (602), Traditional Morality— (604), Law & Order— (605.2), Multiculturalism+ (607), Underprivileged Minorities (705), Demographic Groups (706)

**Right**     Military+ (104), Internationalism— (109), Freedom (201.1), Democracy: General— (202.2), Strong Government (305.3), Pre-Democratic Elites+ (305.4), National Way of Life+ (601), Traditional Morality+ (603), Law & Order+ (605), Civic Mindedness+ (606), Multiculturalism— (608)

**Neutral**    Foreign Special Relationships+/- (101–102), EU+/- (108, 110), Constitutionalism+/- (203–204), Decentralization (301), Centralisation (302), Political Corruption (304), Political Competence (305.1–.2), Bottom-Up Activism (606.2), Middle Class & Professionals (704)

# Stance by Transfer Learning

- **Labels:** Transfer learning from Manifesto Project (sentence-level labels); 12 countries, 9 languages ([Lehmann et al., 2025](#); [Merz et al., 2016](#)).
- **Filter:** Keep only sentences where GPT-4.1 agrees with the human coder ([Burnham et al., 2026](#)) to mitigate low intercoder reliability ([Mikhaylov et al., 2012](#)).
- **Classifier:** fine-tuned multilingual natural language inference (NLI) model (requires small training data) ([Laurer et al., 2024](#); [Burnham et al., 2026](#)).
  - Four tasks: dimension (social or economic), economic stance, social stance, general stance.
  - **Held-out balanced accuracy 0.83** over three classes.

## Stance: Training Labels & Filter

- Stratified sample: up to 1,000 sentences per language–dimension–stance cell; full sentences of  $\geq 8$  words; **48,982 sentences**.
- Every sentence recoded by GPT-4.1 ([OpenAI, 2025](#)). Keep only concordant sentences: a filter, not an adjudication.

| Task |                             | $n$    | Agreement | $\kappa$ |
|------|-----------------------------|--------|-----------|----------|
| T1   | Dimension (Economy/Society) | 48,982 | 78.8%     | 0.58     |
| T2   | Economic pole               | 22,006 | 51.5%     | 0.27     |
| T3   | Social pole                 | 16,586 | 59.4%     | 0.40     |
| T4   | General stance (production) | 48,982 | 52.6%     | 0.28     |

The disagreement is directional: GPT-4.1 recovers 68.2% of the human coder's Left sentences but only 43.5% of the Right ones. Concordant training set: **25,761** stance-labeled sentences.

# Stance: Hypotheses & Training

- **NLI framing:** premise (the sentence) + hypothesis  $\rightarrow$  entailment. Production hypotheses: “This text expresses a {leftist, rightist, neutral} political stance.”
- Four tasks, eleven hypotheses: dimension, economic pole, social pole, general stance.
- **Base model:** bge-m3-zeroshot-v2.0 ([Laurer et al., 2024](#)), binary entailment head.
- Each sentence yields an entailing and a non-entailing pair, in its language and in English translation: **394,087 pairs** from 43,173 sentences.
- Split grouped by *manifesto* (70/15/15): evaluation is on parties and documents never seen in training.
- Early stopping on validation Matthews correlation; one A100 trains it in under an hour.

## Stance: Held-Out Performance

| Task               | $n$   | MCC  | Balanced accuracy |
|--------------------|-------|------|-------------------|
| T1: Dimension      | 6,176 | 0.88 | 0.94              |
| T2: Economic pole  | 1,678 | 0.76 | 0.84              |
| T3: Social pole    | 1,626 | 0.77 | 0.84              |
| T4: General stance | 4,172 | 0.75 | <b>0.83</b>       |

### T4 balanced accuracy by language

|         |      |                   |             |
|---------|------|-------------------|-------------|
| Catalan | 0.86 | French            | 0.81        |
| Swedish | 0.85 | German            | 0.79        |
| Spanish | 0.85 | Galician          | 0.76        |
| Italian | 0.84 | <b>English</b>    | <b>0.82</b> |
|         |      | <b>Portuguese</b> | <b>0.82</b> |

Test manifestos never seen in training. Per-class F1: Left 0.86, Right 0.79, Neutral 0.85.

# Item Response Model

Document  $i$ , topic  $t$ :  $Y_{it}$  = chunk count ( $M_i = \sum_t Y_{it}$ ),  $S_{it} \in [-1, 1]$  = mean stance,  $\theta_i$  = ideal point.

**Stance:**

$$S_{it} \sim \mathcal{N}\left(\alpha_t + \beta_t \theta_i, \frac{\sigma^2}{Y_{it}}\right)$$

**Counts:**

$$Y_i \sim \text{Multinomial}(M_i, \pi_i), \quad \pi_{it} = \exp\{\eta_{it}\} / \sum_s \exp\{\eta_{is}\}$$
$$\eta_{it} = \psi_t + \rho_t \theta_i + \kappa_t(\theta_i^2 - 1)$$

- $\alpha_t, \psi_t$ : topic's baseline lean and baseline salience. Consensus is absorbed here.
- $\beta_t$ : **stance gap**. Which issues split the sides by stance.
- $\rho_t$ : **salience gap**. Which issues each side relatively emphasizes.
- $\kappa_t$ : curvature. Allows for non-monotone salience.

# Identification & Estimation

Location, scale, and reflection are conventions, not findings:

$$\sum_i \theta_i = 0, \quad \frac{1}{T} \sum_t \beta_t^2 = 1, \quad \bar{\theta}_{\text{right}} > 0$$

- **Stage 1:** calibrate parameters from the stance equation  $(\alpha_t, \beta_t, \theta_i, \sigma^2)$ .
- **Stage 2:** jointly fit both equations (Godambe-corrected composite likelihood) ([Godambe, 1960](#); [Varin et al., 2011](#)).
- Composite likelihood  $\rightarrow$  robust standard errors, documents as clusters.

# Estimation: Objective

**Stage 1.** Pilot fit of the stance equation, by alternating least squares:

$$(\hat{\alpha}, \hat{\beta}, \theta^S) = \arg \min \sum_{i, t: Y_{it} > 0} Y_{it} (S_{it} - \alpha_t - \beta_t \theta_i)^2$$

Calibrate and freeze:  $\hat{\sigma}^2$ , an overdispersion factor  $d_i$ , and a Godambe factor  $c_Y$ .

**Stage 2.** One calibrated objective over  $\theta$  and all item parameters:

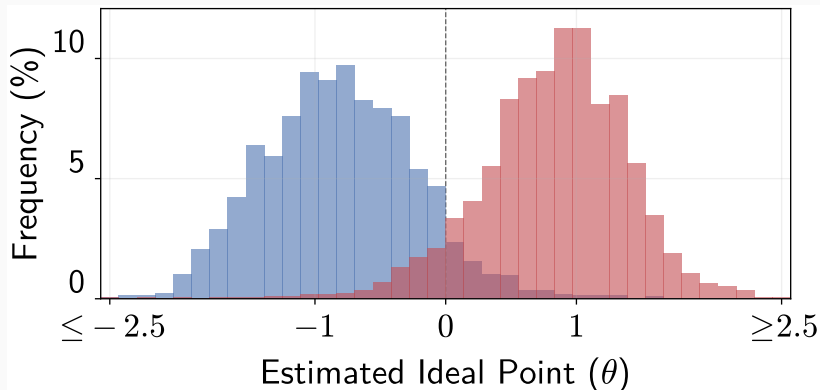
$$\mathcal{L} = \sum_i \left[ \frac{\ell_i^S}{2\hat{\sigma}^2} + \frac{\ell_i^Y}{d_i c_Y} + \frac{\theta_i^2}{2} \right] + \text{pen}(\kappa)$$

- $d_i$  discounts documents that restate the same point: platform text repeats itself.
- $c_Y$  matches the count block's weight to its information about  $\theta_i$ , not its nominal likelihood ([Godambe, 1960](#); [Varin et al., 2011](#)).
- Block coordinate descent; robust SEs from the Godambe information ([White, 1982](#)).

# Centering and Group Means

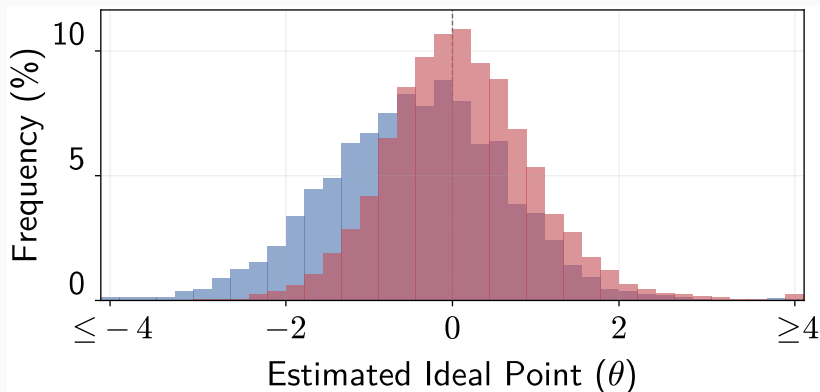
- $\sum_i \theta_i = 0$  is a normalization. With two groups exhausting the sample,  $n_1 \bar{\theta}_1 = -n_2 \bar{\theta}_2$ .
- **US:** 2,187 Democratic and 2,146 Republican platforms; fitted means  $-0.209$  and  $+0.213$ . Their ratio,  $1.019$ , is the ratio of the counts.
- No feature of the text produced that symmetry: either mean determines the other.
- **What carries information:** the separation between groups, the within-group spread, the shape of the distribution. Not the individual means ([Lowe et al., 2011](#); [Clinton et al., 2004](#)).

## Ideal Points: United States



■ Democrats ■ Republicans

## Ideal Points: Brazil



■ Left Parties   ■ Other Parties

# Model Comparison

| Model                                     | United States |             | Brazil       |
|-------------------------------------------|---------------|-------------|--------------|
|                                           | CFScore       | DW-NOMINATE | Expert score |
| <i>Unsupervised scaling, bag-of-words</i> |               |             |              |
| Correspondence analysis, unigrams         | 0.53          | 0.70        | 0.64         |
| Correspondence analysis, bigrams          | 0.65          | 0.74        | 0.54         |
| Wordfish, unigrams                        | 0.72          | 0.80        | 0.19         |
| Wordfish, bigrams                         | 0.76          | 0.81        | 0.73         |
| <i>Document embeddings</i>                |               |             |              |
| Doc2Vec + PCA, unigrams                   | 0.19          | 0.47        | 0.70         |
| Doc2Vec + PCA, bigrams                    | 0.21          | 0.49        | 0.69         |
| Pre-trained Embeddings + PCA              | 0.56          | 0.57        | 0.69         |
| <i>Large language models</i>              |               |             |              |
| Ask and average, GPT-4o mini              | <b>0.87</b>   | <b>0.89</b> | 0.82         |
| Ask and average, GPT-5.4 mini             | 0.85          | 0.88        | 0.82         |
| <i>This paper</i>                         |               |             |              |
| Fasti                                     | 0.81          | 0.85        | <b>0.86</b>  |
| <i>n</i>                                  | 3,788         | 1,111       | 33           |

Pearson correlation of  $\theta$  with each benchmark. Bold: best per column.

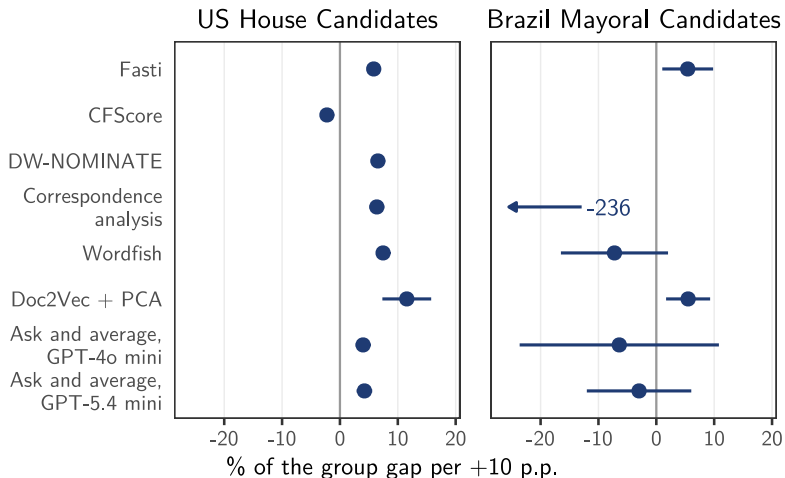
# Model Comparison: Within Party

| Model                                     | CFScore     |             | DW-NOMINATE |             |
|-------------------------------------------|-------------|-------------|-------------|-------------|
|                                           | Dem.        | Rep.        | Dem.        | Rep.        |
| <i>Unsupervised scaling, bag-of-words</i> |             |             |             |             |
| Correspondence analysis, unigrams         | -0.17       | 0.13        | 0.10        | 0.40        |
| Correspondence analysis, bigrams          | -0.17       | <b>0.26</b> | 0.10        | <b>0.41</b> |
| Wordfish, unigrams                        | -0.09       | 0.25        | 0.16        | 0.38        |
| Wordfish, bigrams                         | -0.03       | 0.23        | 0.22        | 0.34        |
| <i>Document embeddings</i>                |             |             |             |             |
| Doc2Vec + PCA, unigrams                   | -0.23       | 0.12        | 0.02        | 0.26        |
| Doc2Vec + PCA, bigrams                    | -0.23       | 0.12        | 0.00        | 0.26        |
| Pre-trained Embeddings + PCA              | -0.08       | 0.23        | -0.02       | 0.38        |
| <i>Large language models</i>              |             |             |             |             |
| Ask and average, GPT-4o mini              | 0.17        | 0.20        | 0.20        | 0.26        |
| Ask and average, GPT-5.4 mini             | <b>0.22</b> | 0.21        | 0.19        | 0.27        |
| <i>This paper</i>                         |             |             |             |             |
| Fasti                                     | 0.10        | 0.20        | <b>0.29</b> | 0.28        |
| <i>n</i>                                  | 2,020       | 1,768       | 576         | 535         |

Bag-of-words and embedding baselines order Democrats against CFScore.

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# Strategic Positioning: All Measures



A row outside the axis is an arrow through the edge, with its estimate.

# Conclusion

- **Main result:** the method recovers interpretable ideal points from text even where party language is *not* polarized (i.e., Brazil).
- **US:** parties use distinct stances and framings, reflecting cultural wars and divergent economic positions.
- **Brazil:** corpus is dominated by support for public services (or neutral). Left: political institutions, minorities, social assistance, values; Right: urban management, technology, infrastructure, public safety, business.
- Within Brazilian parties, candidates vary widely in ideology, yet party means track the national ordering experts perceive.
- Topic + stance outperform embeddings + PCA ([Rheault and Cochrane, 2020](#)); extracting topic and stance from the embeddings greatly improves the estimates.
- No target-domain (US/BR) labels needed (given transfer learning of stance); scalable on modest GPU.